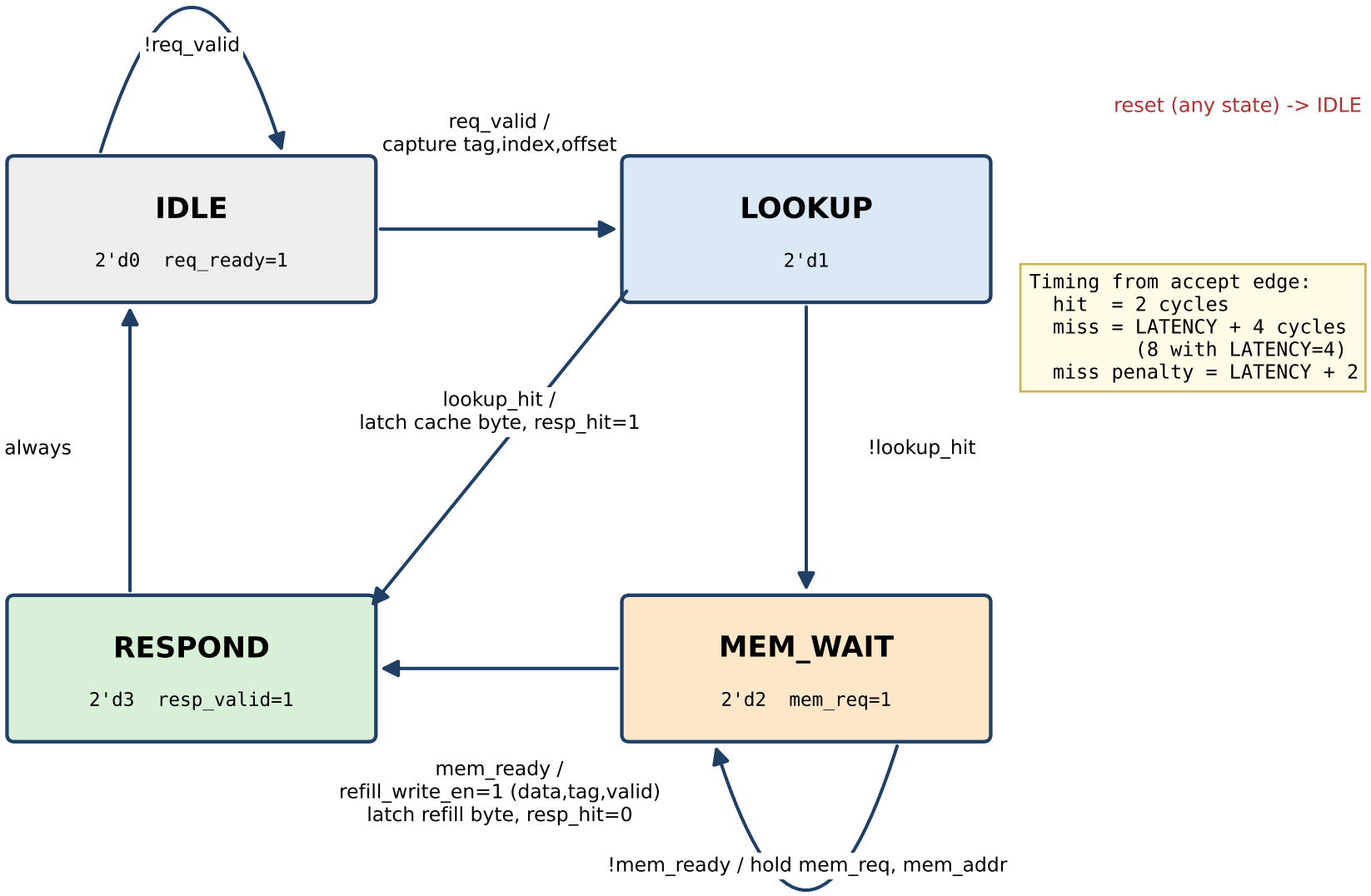
cache_controller.v FSM (as implemented)
!req_valid
IDLE
2'd0 req_ready=1
req_valid /
capture tag,index,offset
LOOKUP
2'd1
reset (any state) -> IDLE
Timing from accept edge:
hit = 2 cycles
miss = LATENCY + 4 cycles
(8 with LATENCY=4)
miss penalty = LATENCY + 2
lookup_hit /
latch cache byte, resp_hit=1
always
!lookup_hit
RESPOND
2'd3 resp_valid=1
MEM_WAIT
2'd2 mem_req=1
mem_ready /
refill_write_en=1 (data,tag,valid)
latch refill byte, resp_hit=0
!mem_ready / hold mem_req, mem_addr